Spring and Spring Boot Interview Questions and Answers (50 Questions)

Topics Covered:

- Spring Core
- Dependency Injection
- Bean Lifecycle
- Spring Boot
- REST API
- Annotations
- Spring MVC
- Spring Data JPA
- Hibernate Basics
- Exception Handling
- Validation
- Profiles
- Configuration
- Microservice Basics
- Security Basics

Suitable for 0–3 years experience.

Spring Core Questions

1. What is Spring Framework?

Answer

Spring Framework is a lightweight Java framework used for building enterprise applications. It provides features like Dependency Injection, Aspect-Oriented Programming, transaction management, and web development support. Spring helps create loosely coupled and maintainable applications.

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2. What are the main modules of Spring?

Answer

Spring provides modules such as Spring Core, Spring MVC, Spring JDBC, Spring ORM, Spring AOP, and Spring Security. Each module focuses on a specific area of application development. Developers can use only the modules they need.

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3. What is Dependency Injection?

Answer

Dependency Injection (DI) is a design pattern where objects receive dependencies from the Spring container instead of creating them manually. DI helps reduce tight coupling and improves testability and maintainability.

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4. What is Inversion of Control (IoC)?

Answer

Inversion of Control means the control of object creation and dependency management is transferred from the application to the Spring container. Spring IoC container manages beans and their lifecycle automatically.

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5. What is a Spring Bean?

Answer

A Spring Bean is an object managed by the Spring IoC container. Beans are created, configured, and maintained by Spring. They are typically defined using annotations or XML configuration.

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6. What is ApplicationContext?

Answer

ApplicationContext is the advanced Spring IoC container responsible for managing beans and application configuration. It provides features like event propagation, dependency injection, and bean lifecycle management.

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7. Difference between BeanFactory and ApplicationContext?

Answer

BeanFactory is the basic IoC container with limited features, while ApplicationContext is more advanced and commonly used. ApplicationContext supports internationalization, events, and automatic bean initialization.

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8. What are stereotype annotations in Spring?

Answer

Stereotype annotations are used to define Spring-managed components. Common stereotypes include @Component, @Service, @Repository, and @Controller. These annotations help Spring automatically detect beans during component scanning.

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9. Difference between @Component, @Service, and @Repository?

Answer

All three are stereotype annotations, but they represent different layers. @Component is generic, @Service is used for business logic, and @Repository is used for database access and exception translation.

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10. What is @Autowired?

Answer

@Autowired is used for automatic dependency injection in Spring. Spring automatically finds and injects matching beans into fields, constructors, or setter methods.

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11. What is constructor injection?

Answer

Constructor injection is a type of dependency injection where dependencies are provided through the constructor. It is preferred because it supports immutability and easier unit testing.

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12. What is bean scope in Spring?

Answer

Bean scope defines the lifecycle and visibility of Spring beans inside the IoC container. It determines how many bean instances Spring creates and how long they live. Common scopes include singleton, prototype, request, session, and application.

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13. What is singleton scope?

Answer

Singleton is the default bean scope in Spring Framework. Only one bean instance is created per Spring container and shared throughout the application. Singleton scope improves memory usage and is commonly used for stateless services.

Answer

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14. What is prototype scope?

Answer

Prototype scope creates a new bean instance every time the bean is requested from the Spring container. Unlike singleton scope, beans are not shared across the application. It is useful when independent object instances are required.

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15. What is bean lifecycle in Spring?

Answer

Spring bean lifecycle includes bean creation, dependency injection, initialization, usage, and destruction. Spring manages these phases automatically using the IoC container. Developers can customize initialization and cleanup using lifecycle callback methods.

Answer

Spring bean lifecycle includes bean creation, dependency injection, initialization, usage, and destruction. Spring provides lifecycle callback methods such as init and destroy.

Spring Boot Questions

16. What is Spring Boot?

Answer

Spring Boot is an extension of Spring Framework used to simplify application development. It provides auto-configuration, embedded servers, starter dependencies, and production-ready features.

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17. What are advantages of Spring Boot?

Answer

Spring Boot reduces boilerplate configuration and speeds up development. It provides embedded servers, starter dependencies, auto-configuration, and easy deployment.

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18. What is auto-configuration in Spring Boot?

Answer

Auto-configuration automatically configures Spring beans based on dependencies available in the classpath. It reduces manual configuration and simplifies setup.

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19. What is @SpringBootApplication?

Answer

@SpringBootApplication is a combination of @Configuration, @EnableAutoConfiguration, and @ComponentScan. It is the main annotation used to start Spring Boot applications.

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20. What is embedded server in Spring Boot?

Answer

Spring Boot provides embedded servers such as Tomcat, Jetty, and Undertow within the application itself. This allows applications to run directly using a main method without external deployment. Embedded servers simplify development and deployment.

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21. What is application.properties?

Answer

application.properties is the main configuration file used in Spring Boot applications. It stores settings such as database configuration, server port, logging levels, and custom application properties. Spring Boot automatically reads this file during startup.

Answer

application.properties is a configuration file used to define application settings such as database connection, server port, and logging properties.

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22. What is application.yml?

Answer

application.yml is an alternative configuration file format supported by Spring Boot using YAML syntax. It provides better readability and supports hierarchical configurations more clearly than properties files. Many enterprise projects prefer YAML for large configurations.

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23. What is Spring Boot Starter?

Answer

Spring Boot Starters are preconfigured dependency packages that simplify project setup and dependency management. They automatically include required libraries for specific functionalities. Examples include spring-boot-starter-web and spring-boot-starter-data-jpa.

Answer

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24. What is DevTools in Spring Boot?

Answer

Spring Boot DevTools improves developer productivity by providing automatic restart, live reload, and enhanced development configuration.

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25. What is CommandLineRunner?

Answer

CommandLineRunner is a Spring Boot interface used to execute code immediately after the application starts. It is commonly used for initialization logic, loading sample data, and testing startup behavior. The run() method executes automatically during startup.

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26. What is Spring MVC?

Answer

Spring MVC is a web framework provided by Spring for building web applications and REST APIs. It follows the Model-View-Controller design pattern to separate business logic, presentation, and request handling. Spring MVC simplifies request processing and web development.

Answer

Spring MVC is a web framework used to build web applications and REST APIs. It follows the Model-View-Controller design pattern.

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27. What is DispatcherServlet?

Answer

DispatcherServlet is the central controller of Spring MVC architecture. It receives all incoming HTTP requests and forwards them to appropriate controllers for processing. It also coordinates view resolution and response generation.

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28. What is @Controller?

Answer

@Controller is a Spring annotation used to define MVC controller classes. Controllers handle incoming requests and typically return view names such as JSP or Thymeleaf pages. It is mainly used in traditional web applications.

Answer

@Controller is used to define Spring MVC controllers that return views such as JSP or Thymeleaf pages.

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29. What is @RestController?

Answer

@RestController is a specialized controller annotation used for REST APIs. It combines @Controller and @ResponseBody.

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30. Difference between @Controller and @RestController?

Answer

@Controller is mainly used for returning views, while @RestController returns JSON or XML responses directly.

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31. What is @RequestMapping?

Answer

@RequestMapping is used to map HTTP requests to controller methods in Spring MVC. It supports multiple HTTP methods such as GET, POST, PUT, and DELETE. Developers can define URL paths and request handling behavior using this annotation.

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32. Difference between @GetMapping and @PostMapping?

Answer

@GetMapping is used to handle HTTP GET requests mainly for retrieving data from the server. @PostMapping is used to handle HTTP POST requests mainly for sending or creating data. Both are shortcut annotations introduced for cleaner REST API development.

Answer

@GetMapping handles HTTP GET requests mainly for retrieving data. @PostMapping handles HTTP POST requests mainly for creating data.

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33. What is @PathVariable?

Answer

@PathVariable is used to extract values from URL paths dynamically.

Example:

```
@GetMapping("/users/{id}")
```

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34. What is @RequestParam?

Answer

@RequestParam is used to read query parameters from HTTP requests in Spring MVC. It extracts values from the URL and maps them to method parameters. It is commonly used for pagination, filtering, and search operations.

Example:

```
/users?page=1
```

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35. What is @RequestBody?

Answer

@RequestBody converts incoming JSON or XML request data into Java objects automatically using Jackson. It is mainly used in REST APIs to receive request payloads. This annotation simplifies request data handling.

Answer

@RequestBody converts incoming JSON request data into Java objects automatically using Jackson.

Spring Data JPA Questions

36. What is Spring Data JPA?

Answer

Spring Data JPA simplifies database access by reducing boilerplate JDBC and Hibernate code. It provides repository interfaces and automatic query generation.

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37. What is JPA?

Answer

JPA stands for Java Persistence API. It is a specification for object-relational mapping and database persistence in Java.

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38. What is Hibernate?

Answer

Hibernate is a popular ORM framework implementing JPA. It maps Java objects to database tables and simplifies database operations.

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39. What is @Entity?

Answer

@Entity is a JPA annotation used to mark a Java class as a database entity. Each entity class is mapped to a database table. Hibernate and JPA use entity classes for ORM operations.

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40. What is @Id?

Answer

@Id is used to define the primary key field of an entity class. The primary key uniquely identifies each record in the database table. It is commonly used along with @GeneratedValue.

Answer

@Id is used to define the primary key field of an entity.

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41. What is JpaRepository?

Answer

JpaRepository is a Spring Data JPA interface that provides built-in CRUD operations, pagination, and sorting support. Developers can perform database operations without writing boilerplate SQL code. It greatly simplifies repository development.

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42. Difference between save() and saveAll()?

Answer

save() is used to persist or update a single entity in the database. saveAll() is used to persist or update multiple entities together in batch style. saveAll() improves performance when handling large datasets.

Answer

save() persists a single entity, while saveAll() persists multiple entities together.

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43. What is pagination in Spring Boot?

Answer

Pagination is used to fetch records in smaller chunks instead of loading all data at once. It improves application performance and reduces memory usage for large datasets. Spring Data JPA provides Pageable and Page interfaces for pagination support.

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44. What is exception handling in Spring Boot?

Answer

Spring Boot provides centralized exception handling mechanisms using @ExceptionHandler and @ControllerAdvice. It helps applications return meaningful error responses instead of exposing internal errors. Centralized handling improves maintainability and consistency.

Answer

Spring Boot provides centralized exception handling using @ExceptionHandler and @ControllerAdvice. It helps return proper error responses.

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45. What is @ControllerAdvice?

Answer

@ControllerAdvice is used for global exception handling across multiple controllers in a Spring Boot application. It centralizes error handling logic and avoids duplicate try-catch blocks in controllers. It improves cleaner API design.

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46. What is validation in Spring Boot?

Answer

Validation ensures incoming request data follows required business rules before processing. Spring Boot commonly uses Bean Validation API annotations such as @NotNull, @Size, and @Email. Validation improves data integrity and prevents invalid requests.

Answer

Validation ensures incoming request data follows required rules before processing. Spring Boot commonly uses Bean Validation API.

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47. What is @Valid?

Answer

@Valid is used to trigger validation on incoming request objects automatically in Spring Boot. It works with validation annotations such as @NotNull and @Size. If validation fails, Spring returns validation error responses.

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48. What is profile in Spring Boot?

Answer

Profiles are used to manage environment-specific configurations such as development, testing, and production. Different profile files can contain separate database URLs and settings. Spring activates profiles using configuration or command-line options.

Answer

Profiles are used to manage environment-specific configurations such as development, testing, and production.

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49. What is actuator in Spring Boot?

Answer

Spring Boot Actuator provides production-ready monitoring and management features for applications. It exposes endpoints such as health, metrics, beans, and application info. Actuator helps monitor application status in real-time.

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50. What is microservice?

Answer

A microservice is a small independent service focused on a specific business functionality. Each microservice can be developed, deployed, and scaled independently. Microservices communicate using REST APIs or messaging systems.

Answer

A microservice is a small independent service focused on a specific business functionality. Microservices communicate through APIs and improve scalability and maintainability.